Milestone 7 - All is Fair in DataGenie

Prospect Company: Trivago

1. Data Sources

Trivago's data ecosystem spans real-time bidding infrastructure, user behaviour tracking, partner integration layers, marketing platforms, and subscription management systems. The following data sources are where DataGenie would be pointed.

1.1 Core Revenue & Auction Systems

CPC Auction Engine Logs This dataset captures all bidding activity from partners such as Booking Holdings, Expedia, and Hotels.com. The data is available across dimensions like country, device type, and accommodation category, and is updated in near real-time. Since partner bids directly influence revenue, this dataset is important for identifying early signals of revenue changes.

Referral Click Tracking Table Every time a user clicks through to a partner booking site, the event is recorded along with the bid price at that moment. This acts as the primary revenue dataset, from which KPIs like CPC Revenue and Revenue Per Qualified Referral (RPQR) can be calculated.

1.2 User Behaviour & Conversion Tracking

Hotel Search Session Table This table captures the full user journey, starting from search to final click-out. It includes steps like results viewed, hotel detail page visits, and whether the user clicked out to a partner. It also contains useful dimensions such as country, device type, accommodation type, and travel preferences. This dataset is mainly used to track conversion rates and identify where users are dropping off in the funnel.

A/B Test Results Store Trivago continuously runs experiments on UI, ranking algorithms, and pricing display. The results of these experiments are stored with details like experiment ID, variant, and performance metrics. This allows KPIs to be analysed across different variants and helps in understanding the impact of product changes.

1.3 Marketing & Spend Data

Marketing Spend Table This dataset contains campaign-level spend across different channels such as TV, paid search, display, and social media. The data is segmented by country and campaign. It is mainly used to evaluate marketing efficiency through metrics like Return on Ad Spend (ROAS).

Branded vs Non-Branded Traffic Table This table separates traffic coming from branded searches and non-branded sources. This distinction is important because branded traffic usually has higher intent, while non-branded traffic reflects acquisition efforts. Monitoring this helps in understanding traffic quality and revenue trends.

1.4 Partner & Advertiser Management

Advertiser Performance Table This dataset provides a partner-level view of performance, including bid values, impressions, and click share. It is useful for identifying dependency on specific partners and tracking any sudden changes in their behaviour that could affect revenue.

1.5 Subscription & Hotelier Platform

Business Studio Subscription Table This table contains information about independent hoteliers subscribed to Trivago's Business Studio. It includes details such as subscription tier, renewal dates, engagement levels, and listing performance. By monitoring this data, it is possible to identify early signs of churn, especially when engagement starts dropping before the renewal period.

2. Key Datasets & KPIs

Based on the identified data sources, the following datasets are the most relevant for DataGenie integration. Each of these datasets directly supports key business metrics and helps in identifying potential risks or performance issues.

2.1 Dataset 1: hotel_referral_clicks

Source: CPC Auction Engine / Data Warehouse

This is the main revenue dataset. Each row represents a qualified referral click, i.e., when a user clicks out to a partner booking platform.

Key Fields: click_id, partner_name, country_code, device_type, accommodation_type, bid_price_eur, is_qualified, click_timestamp

Key KPIs:

CPC Revenue
SUM(bid_price_eur) WHERE is_qualified = TRUE
Represents total revenue generated from referral clicks.

Revenue Per Qualified Referral (RPQR)
SUM(bid_price_eur) / COUNT(*) WHERE is_qualified = TRUE
Shows how much revenue is generated per click.

Partner Bid Share
SUM(CASE WHEN partner_name = 'Booking Holdings' THEN bid_price_eur ELSE 0 END) / SUM(bid_price_eur) * 100
Helps understand dependency on specific partners.

Average CPC by Partner
AVG(bid_price_eur) WHERE is_qualified = TRUE
Useful for tracking changes in partner bidding behaviour.

Dimensions: country_code, device_type, accommodation_type, partner_name

Granularity:

- Daily for revenue metrics
- Hourly for Avg CPC (since bids can change within a day)

2.2 Dataset 2: hotel_search_sessions

Source: User Behaviour / Clickstream

This dataset captures the full user journey from search to click-out.

Key Fields: session_id, country_code, device_type, accommodation_type, search_timestamp, detail_page_viewed, clicked_out, ab_variant

Key KPIs:

Conversion Rate (Search to Click-out)
SUM(CASE WHEN clicked_out = TRUE THEN 1 ELSE 0 END) / COUNT(*) * 100
Measures how effectively users convert.

Detail Page View Rate
SUM(CASE WHEN detail_page_viewed = TRUE THEN 1 ELSE 0 END) / COUNT(*) * 100
Indicates user engagement with listings.

Search Volume
COUNT(*)
Tracks demand and traffic trends.

Result Abandonment Rate
SUM(CASE WHEN clicked_out = FALSE THEN 1 ELSE 0 END) / COUNT(*) * 100
Helps identify drop-offs in the funnel.

Dimensions: country_code, device_type, accommodation_type, ab_variant

Granularity:

- Daily for most metrics
- Hourly during peak periods when faster changes are expected

2.3 Dataset 3: advertiser_bid_activity

Source: Advertiser API / Auction System

This dataset tracks partner bidding behaviour and acts as an early indicator of revenue changes.

Key Fields: partner_name, country_code, device_type, bid_price_eur, bid_timestamp, bid_status

Key KPIs:

Average Partner Bid Price
AVG(bid_price_eur) WHERE bid_status = 'active'
Shows how aggressively partners are bidding.

Bid Reduction Events
COUNT(*) WHERE bid_status = 'reduced'
Helps detect sudden drops in partner participation.

Partner Bid Concentration
SUM(CASE WHEN partner_name = 'Booking Holdings' THEN bid_price_eur ELSE 0 END) / SUM(bid_price_eur) * 100
Indicates dependency on specific partners.

Auction Win Rate
COUNT(clicks_won) / COUNT(auctions_entered) * 100
Measures partner competitiveness in auctions.

Dimensions: partner_name, country_code, device_type

Granularity: Hourly (critical, since bids can change quickly)

2.4 Dataset 4: marketing_spend_performance

Source: Marketing Platforms

This dataset tracks marketing spend and its effectiveness.

Key Fields: country_code, channel, campaign_id, spend_eur, attributed_revenue_eur, branded_flag

Key KPIs:

Return on Ad Spend (ROAS)
SUM(attributed_revenue_eur) / SUM(spend_eur)
Measures efficiency of marketing spend.

Cost Per Acquired Session
SUM(spend_eur) / COUNT(attributed_sessions)
Evaluates cost efficiency of campaigns.

Branded Revenue Share
SUM(CASE WHEN branded_flag = TRUE THEN attributed_revenue_eur ELSE 0 END) / SUM(attributed_revenue_eur) * 100
Shows reliance on high-intent traffic.

Marketing Efficiency Index
SUM(attributed_revenue_eur) / SUM(spend_eur + operational_cost)
Helps evaluate overall marketing effectiveness.

Dimensions: country_code, channel, campaign_id

Granularity:

- Weekly (due to attribution delays)
- Monthly for long-term performance metrics

2.5 Dataset 5: business_studio_subscriptions

Source: Subscription Platform

This dataset supports tracking of recurring revenue and churn.

Key Fields: subscription_id, country_code, subscription_tier, mrr_eur, renewal_date, last_login_date, churn_flag

Key KPIs:

Monthly Recurring Revenue (MRR)
SUM(mrr_eur) WHERE churn_flag = FALSE
Represents stable subscription revenue.

Churn Rate
COUNT() *WHERE churn_flag = TRUE / COUNT()* * 100
Tracks customer loss over time.

Listing Engagement Rate
AVG(click_out_rate_last_30d)
Lower engagement may indicate potential churn.

At-Risk Subscriptions
COUNT(*) WHERE renewal_date BETWEEN TODAY AND TODAY+30
Helps identify subscriptions likely to churn soon.

Dimensions: region, country_code, subscription_tier

Granularity:

- Weekly for churn signals
- Monthly for revenue tracking

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3. Data Quality Problems & Mitigation

Since DataGenie works on time-series data, data quality issues are not just minor inconveniences - they can directly affect the accuracy of anomaly detection. If the underlying data is inconsistent or incomplete, the system may generate misleading insights. One important aspect to note is that DataGenie relies on historical patterns to understand what "normal" looks like. If there are gaps or inconsistencies in this history, it becomes difficult to distinguish between actual anomalies and data issues.

3.1 Missing Time-Series Data If data is missing for a specific day or dimension (for example, no data for Germany on a given day), the baseline for that KPI becomes inaccurate. This can lead to false anomaly detection.

Mitigation: A data completeness check should be implemented before ingestion. Any missing date-dimension combinations should be flagged. Where possible, data should be backfilled from source systems. If data is genuinely unavailable, NULL rows can be inserted with a data quality flag so that such points are excluded from baseline calculations.

3.2 Late-Arriving Data In some cases, revenue data is not final immediately. For example, clicks may be validated later and adjusted, which means recent data may not be fully reliable.

Mitigation: Introduce a delay in considering data as "final". For instance, use T-2 (two days prior) data for KPI calculations, while marking recent data (T-0, T-1) as provisional. This ensures that incomplete data does not lead to incorrect insights.

3.3 Partner Bid Data Gaps If a partner pauses their campaigns, there may be no bid records for that period. This can result in NULL values and affect KPIs like average bid price.

Mitigation: Use COALESCE to handle NULL values in calculations. Additionally, the absence of bids itself can be treated as a signal, and a separate metric such as "partner inactivity" can be tracked.

3.4 Dimension Inconsistency The same dimension (such as country) may appear in different formats across datasets (e.g., "DE", "DEU", "Germany"). This can lead to incorrect grouping and analysis.

Mitigation: Define standard dimension tables (for example, country, device, partner) and map all incoming data to these formats. Any unmatched values should be flagged and handled separately.

3.5 Schema Changes If the structure of the data changes (for example, new fields are added or existing ones are modified), existing KPI calculations may break without immediate visibility.

Mitigation: Implement schema validation checks during data ingestion. Any changes should trigger a review before processing continues. Until resolved, affected KPIs can be temporarily paused to avoid incorrect outputs.

3.6 Marketing Attribution Lag Marketing data, especially from channels like TV, may take a few days to reflect actual revenue impact. Calculating metrics like ROAS immediately can give misleading results.

Mitigation: Use a rolling time window (for example, 7 days) when calculating such KPIs. This accounts for delayed attribution and provides a more accurate view of performance.

4. Cross-Dataset Insights

One of the biggest advantages of DataGenie is that it does not analyse KPIs in isolation. Instead, it connects signals across multiple datasets and presents them as a single, meaningful story. This helps teams quickly understand not just what went wrong, but also why it happened.

Story 1: Revenue Drop with Bid Context

Datasets used: hotel_referral_clicks + advertiser_bid_activity + hotel_search_sessions

- CPC Revenue in the DACH region dropped by 14%
- Partner bid data shows that Booking Holdings reduced bids by 21%
- Search volume actually increased by 9%

Insight: The revenue drop is not due to lower demand, but because of reduced partner bidding.

Why this matters: Without connecting these datasets, the issue could be misinterpreted as a traffic or marketing problem. Instead, the correct action is to address the partner-side issue.

Story 2: Marketing Spend vs Conversion Issue

Datasets used: marketing_spend_performance + hotel_search_sessions + hotel_referral_clicks

- ROAS for TV campaigns in France dropped by 23%
- Search volume increased by 15%
- Conversion rate on mobile dropped by 19%

Insight: Marketing is successfully driving traffic, but the issue lies in conversion, especially on mobile.

Why this matters: This prevents incorrect decisions like reducing marketing spend, and instead points towards fixing the conversion funnel.

Story 3: Early Churn Signal

Datasets used: business_studio_subscriptions + hotel_search_sessions

- Engagement for premium hoteliers dropped by 31%
- Search demand for similar accommodation types dropped by 18%

Insight: The drop in engagement is partly driven by reduced demand, not just platform-related issues.

Why this matters: Helps distinguish between a product problem and a market-driven trend, leading to better intervention strategies.

Story 4: Partner Dependency Risk

Datasets used: advertiser_bid_activity + hotel_referral_clicks

- Booking Holdings' bid share increased to 61%
- Their contribution to revenue increased to 64%

Insight: There is growing dependency on a single partner.

Why this matters: This is a slow-moving risk that might not be visible in daily dashboards, but can have a significant long-term impact if not addressed early.

5. Granularity Strategy

The choice of granularity depends on how quickly changes in a KPI can impact the business. Different datasets require different levels of monitoring.

- **Revenue Data (hotel_referral_clicks):** Tracked daily for overall trends, but hourly monitoring is useful for metrics like CPC where changes happen quickly.
- **User Behaviour (hotel_search_sessions):** Daily tracking is sufficient since conversion patterns are more stable and influenced by daily trends.
- **Partner Bids (advertiser_bid_activity):** Requires hourly monitoring as bid changes can directly impact revenue within a few hours.
- **Marketing Data (marketing_spend_performance):** Best tracked weekly due to attribution delays, especially for channels like TV.
- **Subscriptions (business_studio_subscriptions):** Weekly tracking helps identify churn risks early enough for intervention.

6. Feasibility Verdict

Feasibility: Confirmed

Based on the analysis, Trivago's data ecosystem is well-suited for DataGenie integration. The identified datasets are structured, KPI-ready, and provide sufficient dimensional detail for generating meaningful insights. Most of the potential challenges, such as missing data, delayed updates, and schema changes, can be handled through standard data engineering practices.

Key Consideration

The main challenge lies in data governance. Since Trivago deals with sensitive user and partner data, it is important that DataGenie operates within their existing environment without moving raw data externally.

Recommended approach:

- Connect directly to Trivago's data warehouse using read-only access
- Process and analyse data within their environment
- Expose only aggregated insights instead of raw data

Recommended Starting Point

The best dataset to begin with is hotel_referral_clicks, as it directly represents revenue and has a clean structure. Tracking KPIs like CPC Revenue and RPQR from this dataset can quickly demonstrate value, especially by generating early "top stories" within the first few days of implementation.